BEE 4750 Lab 2: Monte Carlo

Name:

ID:

> **Due Date**
>
> Thursday, 10/02/25, 9:00pm

Setup

The following code should go at the top of most Julia scripts; it will load the local package environment and install any needed packages. You will see this often and shouldn't need to touch it.

```julia
import Pkg
Pkg.activate(".")
Pkg.instantiate()
```

```
  Activating project at `c:\Users\Donny\OneDrive\Post-Grad\Courses\BEE 5750\Lab 2\lab-2-dc226
```

```julia
using Random # random number generation
using Distributions # probability distributions and interface
using Statistics # basic statistical functions, including mean
using Plots # plotting
```

Overview

In this lab, we will conduct a Monte Carlo experiment and explore how sensitive Monte Carlo estimates are to the underlying probability distribution(s).

You should always start any computing with random numbers by setting a "seed," which controls the sequence of numbers which are generated (since these are not *really* random, just "pseudorandom"). In Julia, we do this with the `Random.seed!()` function.

```
Random.seed!(1)
```

```
TaskLocalRNG()
```

It doesn't matter what seed you set, though different seeds might result in slightly different values. But setting a seed means every time your notebook is run, the answer will be the same.

> **Seeds and Reproducing Solutions**
>
> If you don't re-run your code in the same order or if you re-run the same cell repeatedly, you will not get the same solution. If you're working on a specific problem, you might want to re-use `Random.seed()` near any block of code you want to re-evaluate repeatedly.

Probability Distributions and Julia

Julia provides a common interface for probability distributions with the `Distributions.jl` package. The basic workflow for sampling from a distribution is:

1. Set up the distribution. The specific syntax depends on the distribution and what parameters are required, but the general call is the similar. For a normal distribution or a uniform distribution, the syntax is

   ```
   # you don't have to name this "normal_distribution"
   #   is the mean and   is the standard deviation
   normal_distribution = Normal( ,  )
   # a is the upper bound and b is the lower bound; these can be set to +Inf or -Inf for ar
   uniform_distribution = Uniform(a, b)
   ```

   There are lots of both univariate and multivariate distributions, as well as the ability to create your own, but we won't do anything too exotic here.

2. Draw samples. This uses the `rand()` command (which, when used without a distribution, just samples uniformly from the interval $[0, 1]$.) For example, to sample from our normal distribution above:

   ```
   # draw n samples
   rand(normal_distribution, n)
   ```

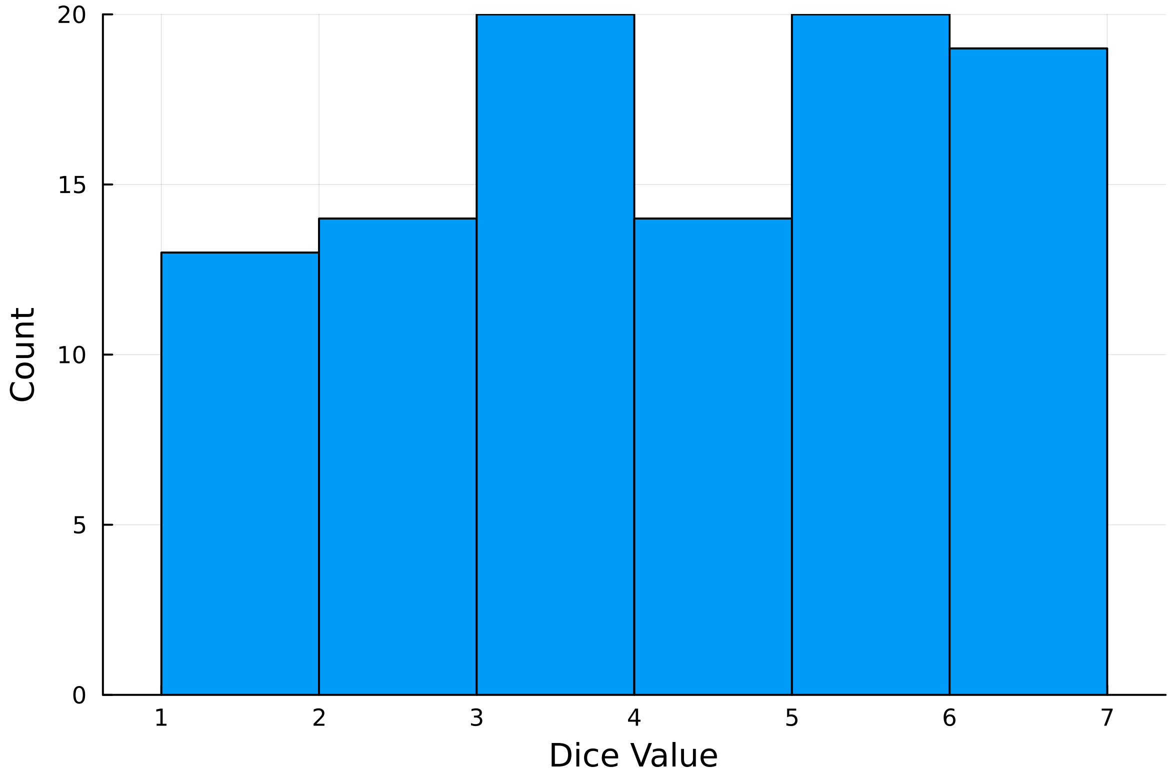
Putting this together, let's say that we wanted to simulate 100 six-sided dice rolls. We could use a Discrete Uniform distribution.

```
dice_dist = DiscreteUniform(1, 6) # can generate any integer between 1 and 6
dice_rolls = rand(dice_dist, 100) # simulate rolls
```

```
100-element Vector{Int64}:
 1
 3
 5
 4
 6
 2
 5
 5
 5
 2

 3
 6
 5
 5
 6
 3
 6
 6
 6
```

And then we can plot a histogram of these rolls:

```
histogram(dice_rolls, legend=:false, bins=6)
ylabel!("Count")
xlabel!("Dice Value")
```



Instructions

Remember to:

- Evaluate all of your code cells, in order (using a `Run All` command). This will make sure all output is visible and that the code cells were evaluated in the correct order.
- Tag each of the problems when you submit to Gradescope; a 10% penalty will be deducted if this is not done.

Problem

A common engineering problem is to quantify flood risk, which is typically computed by propagating a flood hazard distribution through a *depth-damage function* relating flood depths to economic damages. A reasonable depth-damage function for a house without a basement is a bounded logistic function,

$$d(h) = \mathbb{1}_{h>0} \frac{L}{1 + \exp(-k(h - h_0))},$$

where d is the damage as a percent of total structure value, h is the water depth (relative to the house's ground floor elevation) in m, $\mathbb{1}_{x>0}$ is the indicator function, L is the maximum loss in USD, k is the slope of the depth-damage relationship, and h_0 is the inflection point. We'll assume $L = \$200,000$, $k = 0.8$, and $h_0 = 3$.

For this problem, suppose that we have two different probability distributions characterizing annual maxima flood depths:

1. $h \sim LogNormal(1.2, 0.3)$;
2. $h \sim GeneralizedExtremeValue(3, 0.9, -0.15)$.

Plot histograms of samples from these distributions and conduct a Monte Carlo experiment for annual maximum flood damages using each. Answer the following questions:

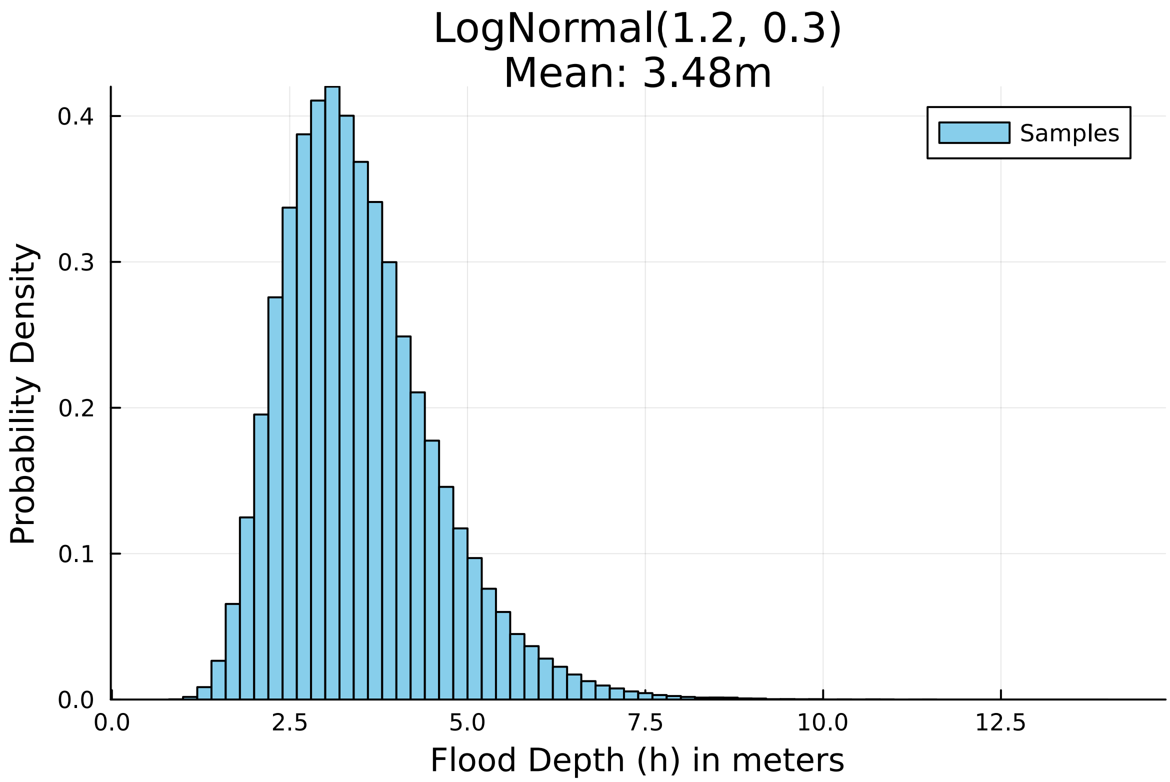
- What are the Monte Carlo estimates of the expected value and the 99% quantile for the annual maximum damage the structure would suffer for each of these flood hazard distributions?
- How did you decide on your sample size?
- Why do you think the estimates differed or did not differ (you can also plot the depth-damage function to compare with the samples)?
- How might you decide (based on getting additional information, if needed) which distribution is "better"?

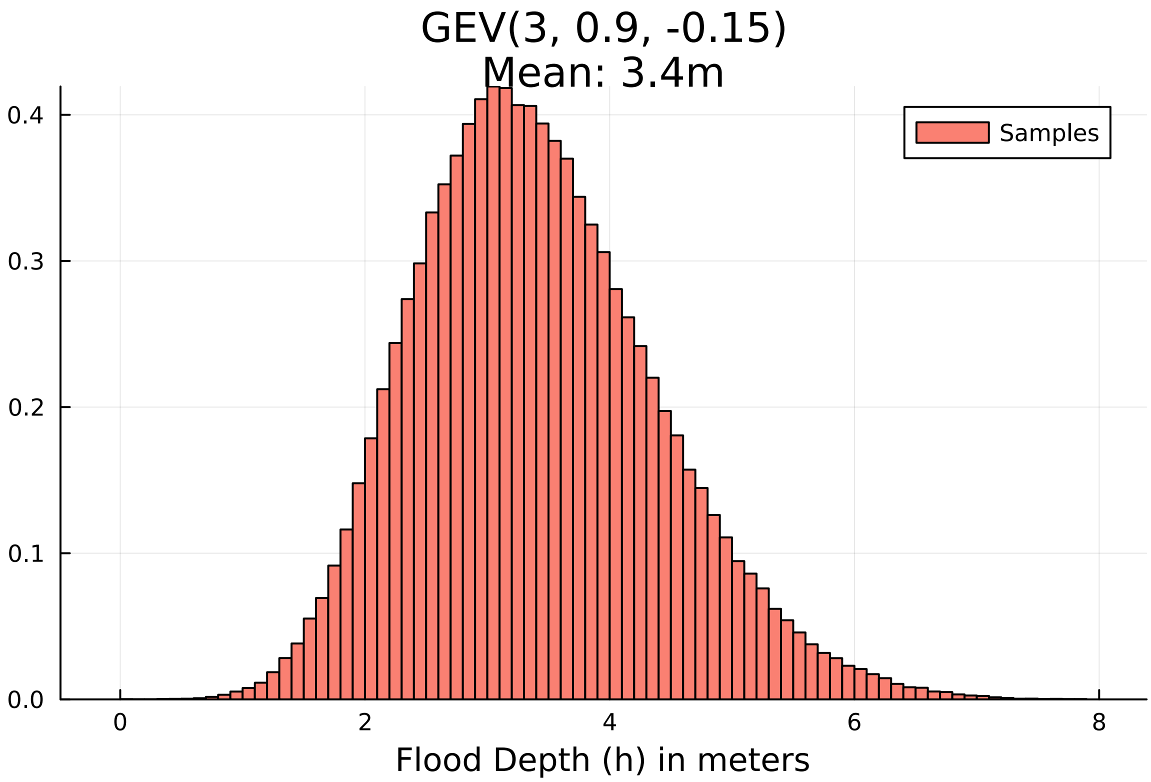
```julia
# define constants
const L = 300_000.0  # Maximum loss in USD (using Float64)
const k = 0.8        # Slope
const h0 = 3.0       # Inflection point
const SAMPLE_SIZE = 200_000

# defining the depth-damage function
"""
    depth_damage_function(h)

Calculates the economic damage for a given flood depth `h` using a
bounded logistic function. Damage is zero for h <= 0.
"""
function depth_damage_function(h::Real)
    if h <= 0
        return 0.0
    end
    return L / (1 + exp(-k * (h - h0)))
end

# create flood depth distribution samples
```

```julia
# lognormal distribution with mean and stddev determined
lognormal_dist = LogNormal(1.2, 0.3)
# GEV distribution with location, scale, shape determined
gev_dist = GeneralizedExtremeValue(3.0, 0.9, -0.15)

# generate random samples
h_lognormal = rand(lognormal_dist, SAMPLE_SIZE)
h_gev = rand(gev_dist, SAMPLE_SIZE)

# create histograms of flood depth distributions

p1 = histogram(h_lognormal,
    bins=100,
    normalize=:pdf,
    label="Samples",
    title="LogNormal(1.2, 0.3)\nMean: $(round(mean(h_lognormal), digits=2))m",
    xlabel="Flood Depth (h) in meters",
    ylabel="Probability Density",
    color=:skyblue,
    legend=:topright
)
display(p1)

p2 = histogram(h_gev,
    bins=100,
    normalize=:pdf,
    label="Samples",
    title="GEV(3, 0.9, -0.15)\nMean: $(round(mean(h_gev), digits=2))m",
    xlabel="Flood Depth (h) in meters",
    color=:salmon,
    legend=:topright
)
display(p2)


# monte carlo simulation of damages

# distributing flood depths through the depth-damage function
damage_lognormal = depth_damage_function.(h_lognormal)
damage_gev = depth_damage_function.(h_gev)

# finding data requested
```

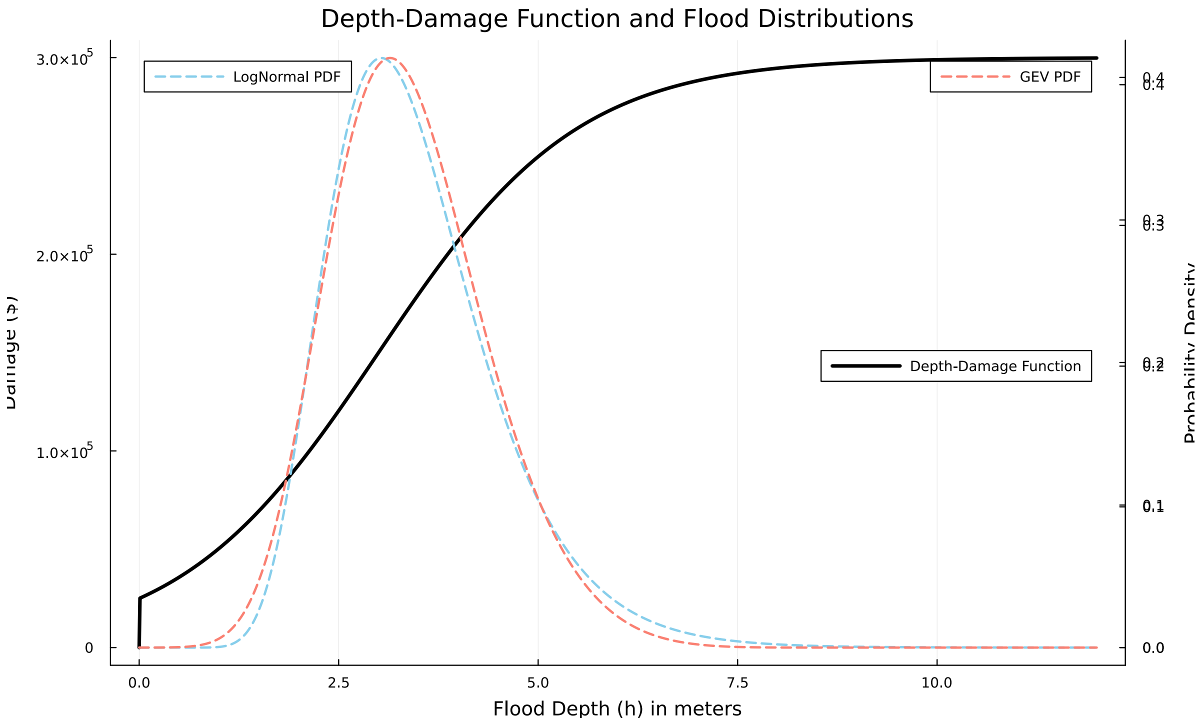
```
# expected value of damage
expected_damage_ln = mean(damage_lognormal)
expected_damage_gev = mean(damage_gev)

# 99% quantile of damage
quantile_99_ln = quantile(damage_lognormal, 0.99)
quantile_99_gev = quantile(damage_gev, 0.99)

# print results
println("Sample Size for each distribution: ", SAMPLE_SIZE, "\n")

println("LogNormal Distribution")
println("Estimated Expected Annual Damage: \$", expected_damage_ln)
println("Estimated 99% Quantile of Damage: \$", quantile_99_ln, "\n")

println("Generalized Extreme Value (GEV) Distribution")
println("Estimated Expected Annual Damage: \$", expected_damage_gev)
println("Estimated 99% Quantile of Damage: \$", quantile_99_gev, "\n")


# --- 7. Explain Differences with a Combined Plot ---

# Create a range of h values for plotting the functions
h_vals = 0:0.01:12
damage_vals = depth_damage_function.(h_vals)
pdf_lognormal = pdf.(lognormal_dist, h_vals)
pdf_gev = pdf.(gev_dist, h_vals)

# Create the plot with a secondary y-axis
plot(h_vals, damage_vals,
    label="Depth-Damage Function",
    xlabel="Flood Depth (h) in meters",
    ylabel="Damage (\$)",
    color=:black,
    lw=3,
    legend=:right,
    size=(1000, 600),
    title="Depth-Damage Function and Flood Distributions"
)

# Plot PDFs on the secondary axis
plot!(twinx(), h_vals, pdf_lognormal,
```

```
    label="LogNormal PDF",
    ylabel="Probability Density",
    color=:skyblue,
    linestyle=:dash,
    lw=2,
    legend=:topleft
)
plot!(twinx(), h_vals, pdf_gev,
    label="GEV PDF",
    color=:salmon,
    linestyle=:dash,
    lw=2
) |> display
```






```
WARNING: redefinition of constant Main.L. This may fail, cause incorrect answers, or produce
```




```
Sample Size for each distribution: 200000

LogNormal Distribution
Estimated Expected Annual Damage: $172831.28287796286
Estimated 99% Quantile of Damage: $284826.88036480703

Generalized Extreme Value (GEV) Distribution
Estimated Expected Annual Damage: $170365.7104879747
Estimated 99% Quantile of Damage: $275219.6296545093
```

References

Put any consulted sources here, including classmates you worked with/who helped you.